



Computer Aided Engineering Graphics (25CAG101)

A.Y:2025-2026

I-B.Tech I-Sem I-Mid Question Bank with Answers

Branch: CSE/CSD

2 Mark Questions with Answers

1. Define Engineering Drawing.

Solution:

Engineering Drawing is a graphical representation of an object, idea, or system that conveys detailed information for design, manufacture, and construction.

It uses standard symbols, dimensions, and scales to communicate shape, size, and features of an object accurately.

2. List out various the principles and significance of Engineering Drawing.

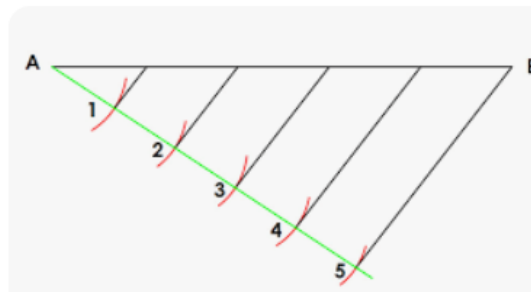
Solution: Principles of Engineering Drawing:

1. **Accuracy:** Drawings must represent the true shape and size of the object.
2. **Clarity:** Lines, symbols, and dimensions should be neat and easy to read.
3. **Simplicity:** Use standard methods and avoid unnecessary details.
4. **Standardization:** Follow BIS or ISO standards for lines, symbols, and dimensions.

3. Divide a straight line 72 mm into a given number of equal parts, say 5

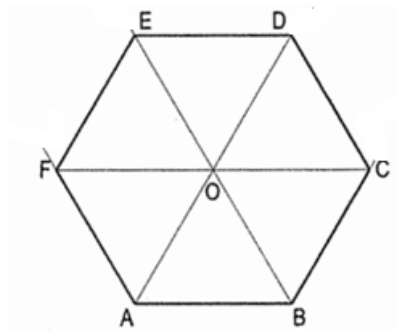
Solution: Draw a straight line AB = 72 mm.

→ Now, AB is divided into 5 equal parts.



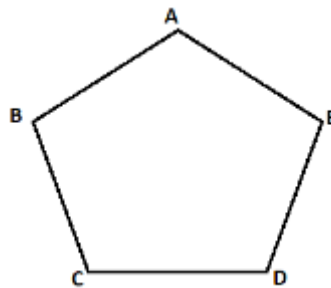
4. Draw a hexagon with a given side 30 mm

Solution:



5. Draw a pentagon with a given side 30 mm

Solution:



6. List various Draw commands used in AutoCAD

Solution:

- Line – Draws straight lines.
- Circle – Draws circles.
- Arc – Draws arcs.
- Polygon – Draws polygons.
- Ellipse – Draws ellipses.

7. List out various Modify commands in improving drawing accuracy

Solution:

- Move – Shifts objects from one place to another.
- Copy – Duplicates selected objects.
- Rotate – Rotates objects at a specified angle.
- Mirror – Creates a mirrored copy.
- Scale – Changes the size proportionally.

8. List out and explain various Dimensional Commands used in AutoCAD.

Solutions: Common Dimensional Commands:

- i. DIMLINEAR
- ii. DIMALIGNED
- iii. DIMRADIUS .

- iv. DIMDIAMETER .
- v. DIMANGULAR

9. Which method is used to construct an ellipse in the general method?

Solution: Eccentricity method using foci and directrix

10. Define eccentricity.

Solution:

Eccentricity (e) is the ratio of the distance of any point on a conic from its **focus** to its perpendicular distance from the **directrix**.

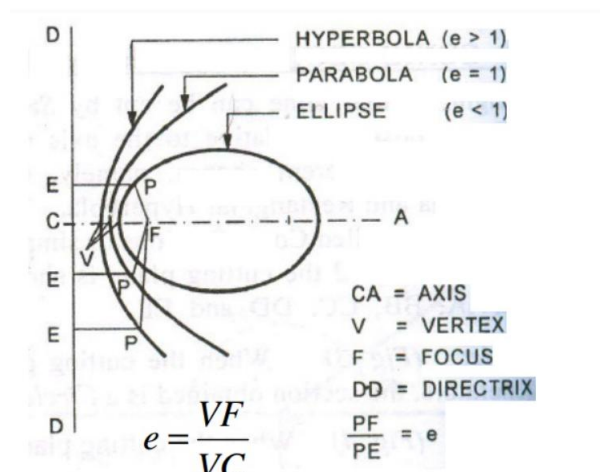
$$e = \frac{\text{Distance from Focus}}{\text{Distance from Directrix}}$$

11. Differentiate between ellipse, parabola and hyperbola based on eccentricity

Solutions:

Curve	Eccentricity (e)	Nature of Curve
Ellipse	$e < 1$	Closed curve
Parabola	$e = 1$	Open curve (one branch)
Hyperbola	$e > 1$	Open curve (two branches)

12. Define eccentricity in graphical form and write the e value for ellipse, parabola and hyperbola



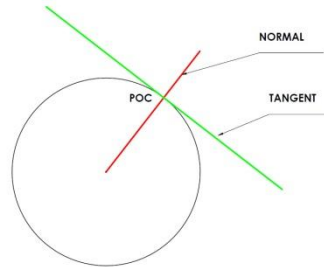
13. What is tangent and normal?

Tangent:

- A tangent is a straight line that touches a curve at one point only. It represents the direction of the curve at that point.

Normal:

- A normal is a straight line drawn perpendicular to the tangent at the point of contact. It shows the direction of the radius or line of symmetry at that point.



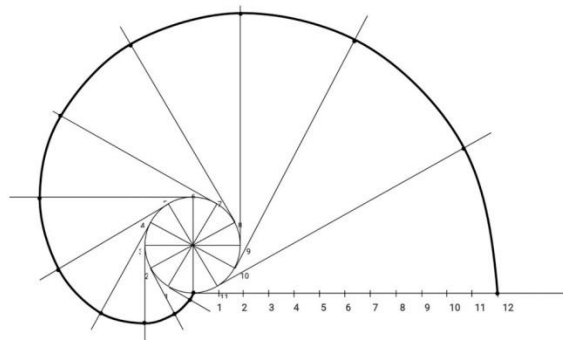
Example: For a circle, the radius at the point of contact is always the normal, and the tangent is perpendicular to it.

14. Define Involute.

Solution: An Involute is a curve traced by the free end of a thread unwound from a circle or a polygon in such a way that the thread is always tight and tangential to the circle or side of the polygon.

15. How can an involute be drawn from a base circle?

Solution:

INVOLUTE OF A CIRCLE**16. Name one practical application of an involute curve**

Solution:

- Used in gear tooth profiles
- In cam design
- In string unwinding mechanisms
- In engineering curves for motion transmission with constant velocity ratio.

17. Name the two principal planes used in orthographic projection

Solution:

- Horizontal Plane (HP)
- Vertical Plane (VP)

18. What is orthographic projection?

Solution Orthographic projection is a method of representing a 3D object on 2D planes (HP and VP) using perpendicular projectors.

It shows true shape and size of each face.

19. In which quadrant is a point located if it is above HP and in front of VP?

Solution: When a point is above the Horizontal Plane (HP) → it lies in the upper region.

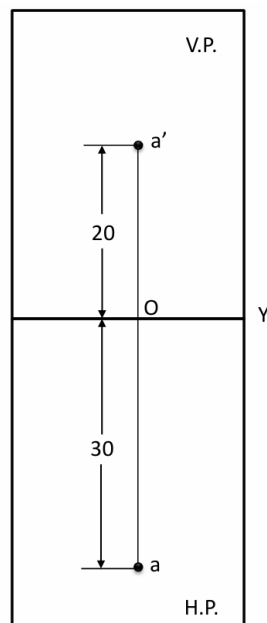
When a point is in front of the Vertical Plane (VP) → it lies toward the observer.

Hence, such a point lies in the First Quadrant of the projection system.

→ **The point lies in the First Quadrant.**

20. Point A is 20 mm above H.P. and 30 mm in front of V.P. draw its front view and top view

Solution:

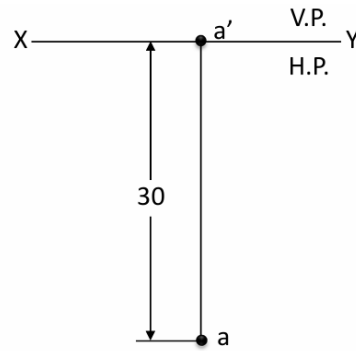


a'oa' : Projector
a' : View from front / Front View
a : View from top / Top View

21. Draw the projections of a point A lying on H.P. and 30 mm in front of V.P. Draw its projections

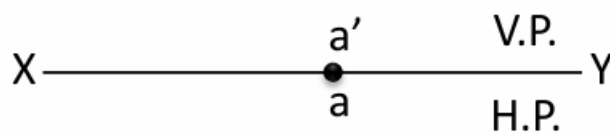
Solution:

a'oa' : Projector
a' : View from front / Front View
a : View from top / Top View



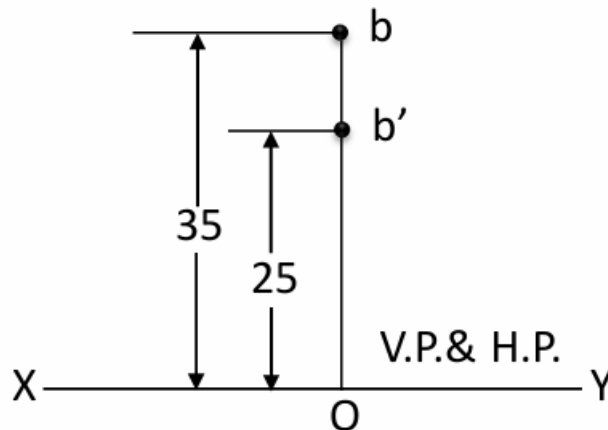
22. Draw the projections of a point F which lies in both the H.P. and V.P. Draw its projections

Solution:



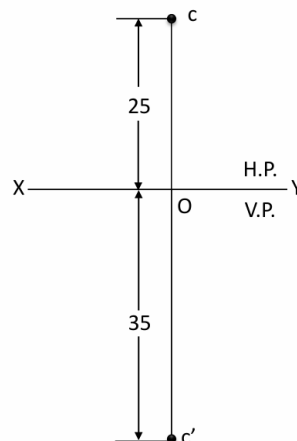
23. A point B is 25 mm above H.P. and 35 mm behind V.P. Draw its projections.

Solution:



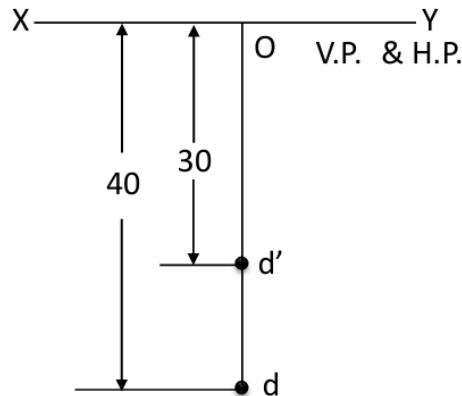
24. A point C is 35 mm below H.P. and 25 mm behind V.P. Draw its projections

Solution:



25. A point D is 45 mm below H.P. and 55 mm behind V.P. Draw its projections.

Solution:



5 Mark Questions with Answers

1. Explain the significance of engineering drawing in the field of engineering

Solution:

1. Accuracy

Definition: Drawings must precisely represent the shape, size, and details of the object.

Significance: Ensures components are manufactured correctly.

2. Legibility

Definition: Drawings should be clear and easy to read.

Significance: Minimizes misinterpretation by machinists and engineers.

3. Clarity and Simplicity

Definition: Use of simple language, standard symbols, and neat layout.

Significance: Facilitates better understanding even for non-technical users.

4. Standardization

Definition: Adhering to recognized standards like BIS or ISO.

Significance: Enables consistency and universal understanding.

5. Scale

Definition: Drawings are made proportional using a defined ratio.

Significance: Helps represent large or tiny components on paper.

6. Projection Methods

Definition: Techniques like orthographic or isometric projections.

Significance: Provides multiple views to fully describe an object.

7. Dimensioning

Definition: Indicating size and location of features.

Significance: Essential for manufacturing and inspection.

8. Line Types and Symbols

Definition: Usage of visible, hidden, center lines, and standard symbols.

Significance: Differentiates object features clearly.

2. Explain types of dimensions in engineering drawing along with examples

Solution:

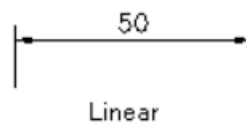
Dimensions are used in engineering drawings to specify the **size, shape, and location** of an object or feature.

They are mainly classified into the follow

1. Linear Dimensions:

- Indicate the **length, width, or height** of an object.
- Shown by dimension lines with arrowheads and numerical values.

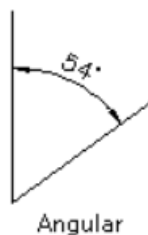
Example: Length of a line = 50 mm



2. Angular Dimensions:

- Specify the **angle** between two lines or surfaces.
- Represented by an arc with the angle value.

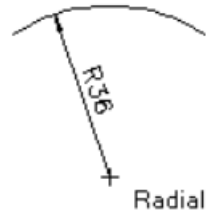
Example: Angle between two inclined lines: 54°



3. Radial Dimensions:

- Used to show the **radius of arcs, circles, or fillets.**
- The dimension line is drawn from the center to the arc.

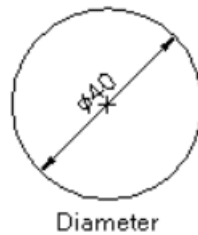
Example: Radius of a circle $R = 36$ mm.



4. Diameter:

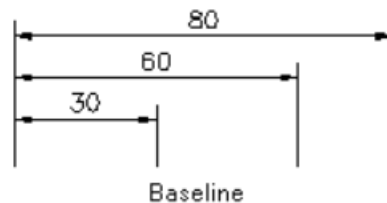
- Used to specify the **diameter of circles or cylindrical objects**.
- Indicated by a symbol “ $\varnothing$ ”.

Example: $\varnothing 40$ mm hole or shaft.



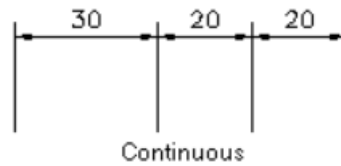
5. Baseline Dimensioning (Parallel Dimensioning):

- In this method, all dimensions are measured from a **common reference line (baseline)**.
 - The baseline may be an edge, center line, or datum.
 - It ensures **accuracy** because all measurements originate from the same point, reducing cumulative errors.



6. Continuous Dimensioning (Chain Dimensioning):

- In this method, dimensions are given **one after another in a chain** from point to point.
- Each dimension starts where the previous one ends.
- Simple to draw but **errors may accumulate** along the chain.



3. **Explain the methods of dimensioning in Engineering Drawing along with examples.**

Solution:

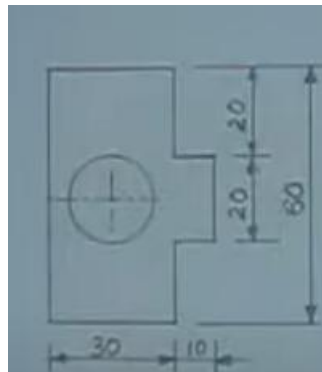
Dimensioning is the process of indicating the **size, location, and other details** of an object on a drawing.

There are mainly **two methods of dimensioning** used in engineering drawing:

Aligned Method

- In this method, the **dimension figures are placed parallel to the dimension lines.**
- They are **read from the bottom or right-hand side** of the drawing sheet.
- The dimension line is drawn parallel to the feature being dimensioned.
- Commonly used in **architectural and machine drawings.**

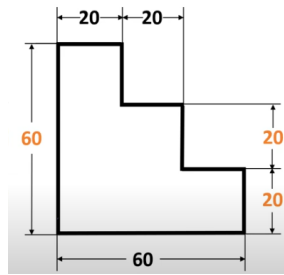
Example:



Unidirectional Method

- In this method, **all dimension figures are written horizontally** for easy reading from the bottom of the sheet.
- Dimension lines may be horizontal, vertical, or inclined, but the figures remain horizontal.
- This method provides a **neat and uniform appearance.**
- Commonly used in **engineering and mechanical drawings.**

Example:



4. What are the advantages of CAD over manual drafting?

Solution:

CAD (Computer-Aided Design) offers many advantages over traditional **manual drafting** methods. It improves speed, accuracy, efficiency, and flexibility in the design process.

1. **High Accuracy:** CAD ensures precise measurements and eliminates human error in drawing.
2. **Time Saving:** Drawings can be created and modified quickly compared to manual drafting.
3. **Easy Editing:** Any changes can be made easily using commands like Move, Trim, and Copy without redrawing.
4. **Reusability of Drawings:** Existing drawings can be reused for new projects, saving time and effort.
5. **Better Visualization:** CAD allows 2D and 3D modeling for realistic visualization of objects.
6. **Improved Drawing Quality:** Drawings are neat, standardized, and consistent in appearance.
7. **Easy Storage and Retrieval:** Digital files can be stored, shared, and retrieved easily without physical space.
8. **Layer Management:** Layers help organize different parts of a drawing (dimensions, outlines, etc.) efficiently.
9. **Easy Communication:** CAD files can be shared electronically for team collaboration and review.
10. **Integration with Other Software:** CAD integrates with CAM and CAE for analysis, simulation, and manufacturing.

5. Explain the basic steps to set up the AutoCAD workspace for a 2D drawing.

Solution: To set up AutoCAD for 2D drawing, follow these basic steps:

1. **Open AutoCAD:** Launch AutoCAD and select a **2D Drafting & Annotation workspace**.

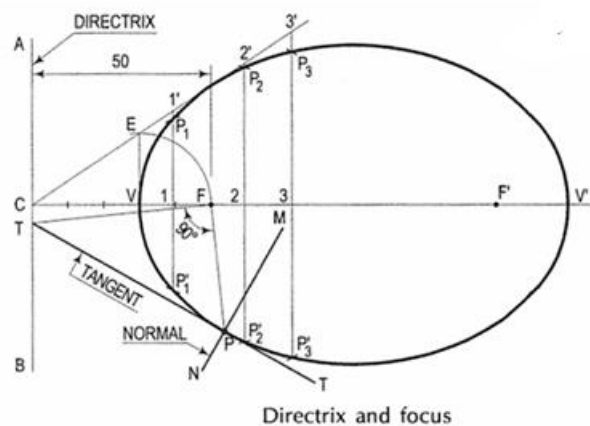
2. **Set Units:** Use the UNITS command to choose the **drawing units** (mm, cm, inches, etc.) and precision.
3. **Set Limits and Grid:**
 - Define the **drawing limits** using the LIMITS command (e.g., 0,0 to 297,210).
 - Turn on the **grid** for guidance (GRID command) and snap to grid (SNAP command) for accuracy.
4. **Set Drawing Scale:** Determine the scale for the drawing to ensure all objects fit properly on the layout.
5. **Layers Setup:**
 - Create and organize **layers** for different components (e.g., dimensions, outlines, centerlines) using the LAYER command.
 - Assign different **colors, line types, and line weights** to layers for clarity.
6. **Snap and Ortho Settings:**
 - Enable **Ortho mode (F8)** for horizontal and vertical lines.
 - Use **Object Snap (OSNAP, F3)** to accurately pick points on objects.
7. **Save Template (Optional):** Save the workspace as a **template file** for future 2D drawings to maintain consistency.

6. Compare the manual drawing with AutoCAD.

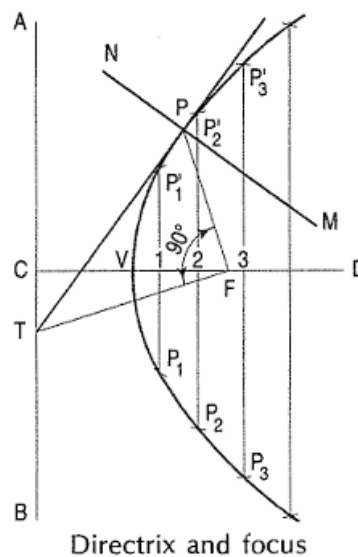
Feature	Manual Drawing	AutoCAD
Accuracy	Less accurate; prone to human error	Highly accurate; precise dimensions and angles
Time	Time-consuming	Fast and efficient
Editing/Modification	Difficult; requires erasing and redrawing	Easy; objects can be moved, copied, or modified quickly
Storage	Requires physical space; hard to store and manage	Digital storage; easy to save, retrieve, and share
Reusability	Low; drawings cannot be reused easily	High; drawings can be reused for future projects
Visualization	2D only; hard to visualize complex shapes	2D and 3D modeling possible
Drawing Quality	Dependent on drafter's skill; may vary	Neat, standardized, and consistent
Collaboration	Difficult to share and work	Easy to share and collaborate

Feature	Manual Drawing	AutoCAD
	simultaneously	electronically
Integration with Other Tools	Not possible	Can integrate with CAM, CAE, and other software
Cost	Low initial cost; needs drawing tools and paper	Higher initial cost; requires software and computer

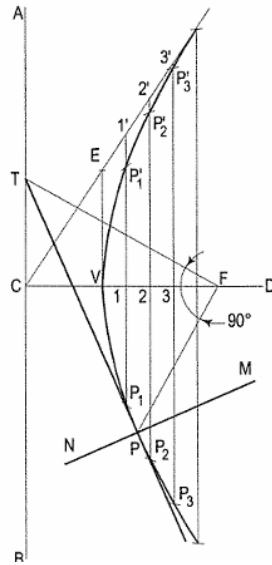
7. Draw an ellipse with the distance of the focus from the directrix is 50 mm and Eccentricity= $\frac{2}{3}$ Also draw a tangent and a normal to the ellipse at a point 60 mm from the directrix. (Eccentricity method)



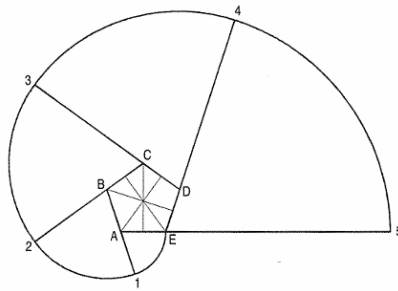
8. Draw a parabola with the distance of the focus from the directrix is 50 mm. Also, draw a tangent and a normal to the parabola at a point on the curve.



9. Draw a hyperbola with the distance of the focus from the directrix = 65 mm and $e = \frac{3}{2}$. Also, draw a tangent and a normal to the curve at a point 45 mm from the directrix.

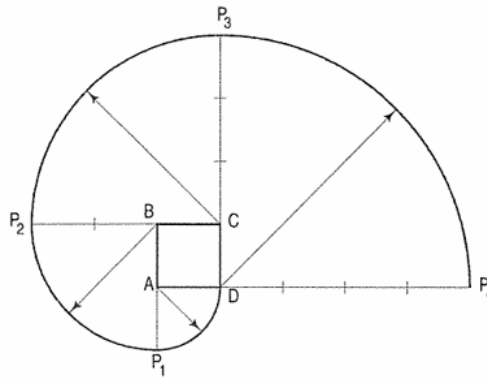


10. Draw an involute of a pentagonal plane of given side 30 mm



11. Draw Hexagonal Involute of given side 25mm

12. Draw a square involute of a given side 40 mm



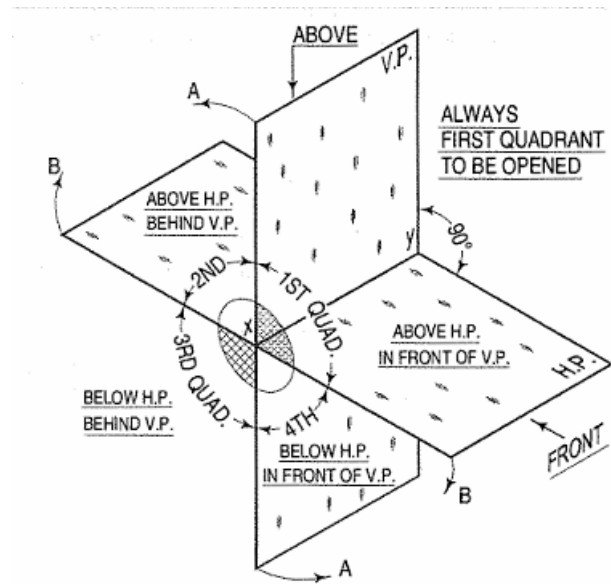
13. a) Draw the projections of a point A located 40 mm above HP and 30 mm in front of VP.

b) Draw projections of a point B 20 mm below HP and 25 mm behind VP

14. a) Draw the projections of a point X located 40 mm below HP and 30 mm in front of VP

b) Draw projections of a point Y 20 mm above HP and 25 mm behind VP.

15. Name the position of a point if it is situated in I, II, III & IV quadrants.



- **HP (Horizontal Plane):** Represents top view.
- **VP (Vertical Plane):** Represents front view.
- **Quadrant I:** Above HP, in front of VP (most commonly used for projections).
- **Quadrant II:** Above HP, behind VP.
- **Quadrant III:** Below HP, behind VP.
- **Quadrant IV:** Below HP, in front of VP.